Extracting Social and Clinical Impacts of Substance Use from Reddit with Structured Pipelines and Ensemble Search

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Abstract

We present our submission to the SMM4H–HeaRD 2026 Task 7 shared task, which asks systems to extract *Clinical* and *Social* impacts of substance use from Reddit posts as a token-level BIO tagging problem. The benchmark is difficult for two reasons: the label prior flips between train and test (train is ≈75% Clinical, test is ≈70% Social), and many entities are implicit colloquialisms without medical vocabulary. We compare five model families—BiLSTM–CRF, GLiNER (fine-tuned), SocBERT, Stress-RoBERTa, and DeBERTa-v3-large—under a consistent learning-rate sweep, and organise our contributions into seven buckets: classical loss and prompt ablations, advanced regularisation, a *preprocessing axis* treating raw and cleaned Reddit text as two experimental regimes, four structured-prediction pipelines, domain-adapted backbones, weight-space model soup, and an exhaustive prediction-space ensemble search. Our best single model is definition prompting at relaxed F1 $= 0.613$. Our best overall system is a four-model majority-vote ensemble at relaxed F1 $= 0.635$, beating the best published baseline on the same dev split by $+0.025$. The main lesson is that training-objective diversity, combined with hard majority voting, is a stronger lever than polishing any single model.

1 Introduction

Public-health surveillance on social media has become an important source of population-level signal for substance-use disorders (Sarker et al., 2019). The SMM4H–HeaRD 2026 shared task Track 7 asks systems to extract, from opioid-subreddit posts, two kinds of consequences that users describe in natural language: *ClinicalImpacts* (physiological and psychological effects such as overdose or withdrawal) and *SocialImpacts* (occupational, legal, and relational consequences such as losing a job or getting arrested). The task is formulated as token-level NER with five BIO tags: `O`, `B/I-ClinicalImpacts`, `B/I-SocialImpacts`.

What makes this benchmark unusually difficult is a striking mismatch between the training and test distributions. The training split is dominated by Clinical entities (≈75%) and the test split flips to Social (≈70%). On top of that, many impacts are expressed *implicitly*—"those weeks were hell" is a Clinical span without a single clinical token—so lexical and ontology-based cues are weak. The published baseline on this version of the dataset reaches relaxed F1 $= 0.610$ and human annotator agreement is around 0.81 (Garg et al., 2025), leaving a $+0.20$ headroom that no single well-known recipe closes.

We treat this as a systems problem and report a thorough comparison across five model families under a uniform LR sweep and a paired *raw vs preprocessed* protocol. Our contributions are:

1. A disciplined baseline study across BiLSTM–CRF, GLiNER fine-tuned, SocBERT, Stress-RoBERTa, and DeBERTa-v3-large, showing that a general-purpose encoder dominates Reddit- and mental-health-adapted encoders on this task (§5.1).
2. A *preprocessing-as-axis* design in which every single model is run in both a raw-text regime and a mojibake/URL/emoji-cleaned regime, so preprocessing is a first-class experimental variable rather than a confound (§4).
3. Seven contribution buckets on top of DeBERTa: focal loss, definition prompting, multi-task learning, recall-boost O-downweighting, R-Drop, FGM + SWA, structured prediction pipelines, weight-space model soup, and prediction-space ensemble search (§5).

4. An exhaustive ensemble search across both preprocessing regimes. A four-model majority-vote ensemble reaches relaxed F1 = **0.635** on the dev set, exceeding the published baseline by $+0.025$ on the same split (§5.2.7).

The clearest single takeaway from the study is that *training-objective diversity* is worth more than squeezing an additional point out of any individual recipe. Five of the seven individual innovations sit at or below the plain DeBERTa baseline when measured alone; they earn their keep by producing decorrelated errors that hard voting combines.

2 Related Work

Health-focused NER on social media. Extracting clinically-relevant spans from social-media text has been studied for medication mentions, adverse drug events, and more recently substance-use impacts (Sarker et al., 2019; Weissenbacher et al., 2022; Garg et al., 2025). The dominant approach is a transformer encoder with a token-classification head, sometimes augmented with domain adaptation. Reddit-Impacts 2.0 (Garg et al., 2025) is unusual in separating *clinical* and *social* outcomes and in deliberately flipping the label prior between train and test.

Domain-adapted encoders. Reddit-specific pre-training (Ji et al., 2022) and mental-health-domain pre-training (Amalq, 2022) typically help when the downstream task shares vocabulary and register with the pre-training corpus. In our setting, however, the task labels are not register-bound and generic DeBERTa-v3 (He et al., 2021) outperforms both domain-adapted encoders.

Span-based NER. GLiNER (Zaratiana et al., 2024) frames NER as open-type span scoring and is attractive for low-resource settings. We include both zero-shot and fine-tuned variants, and a span-nested decoder that scores nested candidates explicitly.

Regularisation and ensembling. Our advanced-regularisation bucket is built from off-the-shelf ingredients: focal loss (Lin et al., 2017), R-Drop consistency (Liang et al., 2021), FGM adversarial training (Miyato et al., 2017), and SWA (Izmailov et al., 2018). Model soup (Wortsman et al., 2022) and prediction-space ensembles (Kuncheva and Whitaker, 2003) are well-established; our contribution is the exhaustive cross-regime subset search.

3 Task and Dataset

Reddit-Impacts 2.0. The dataset contains 1,378 posts from opioid-related subreddits, pre-split into train, dev, and test. The training and dev splits are both Clinical-heavy, while the test split flips to roughly 70% Social. Figure 1 visualises the split composition, the span-length distribution, and the fraction of posts that contain no entities at all (a non-trivial class, particularly on dev).

Evaluation. We report both Relaxed and Strict F1 over entity mentions. Relaxed F1 credits a prediction whose type is correct and whose span overlaps the gold span by at least one token; Strict F1 additionally requires exact span boundaries. Dev-set performance is the primary selection criterion because test-set labels are not available; we discuss the implications of this in §9.

4 Preprocessing as an Axis

A common pitfall in social-media NER is to conflate preprocessing with modelling: a single "cleaned" pipeline is used for every experiment and its effect is invisible. We treat preprocessing as a first-class experimental axis. Every family is run twice:

- **Raw regime.** Original Reddit text goes straight to the tokeniser. Usernames, URLs, mojibake (broken Unicode from earlier encoding/decoding trips), emoji, and stray punctuation are preserved. This regime maximises lexical fidelity but produces noisy token boundaries.
- **Preprocessed regime.** We repair mojibake, mask `@user` mentions and URLs with placeholder tokens, strip emoji and long punctuation runs, and restore masked positions as `O` tags at inference time so that the BIO sequence length is preserved.

The effect of the regime is family-specific. BiLSTM–CRF and GLiNER both gain markedly from preprocessing (relaxed F1 $+0.045$ and $+0.061$ respectively). The plain DeBERTa baseline is actually *stronger* on raw text (0.595 vs 0.545), which is consistent with the hypothesis that large pre-trained encoders are already robust

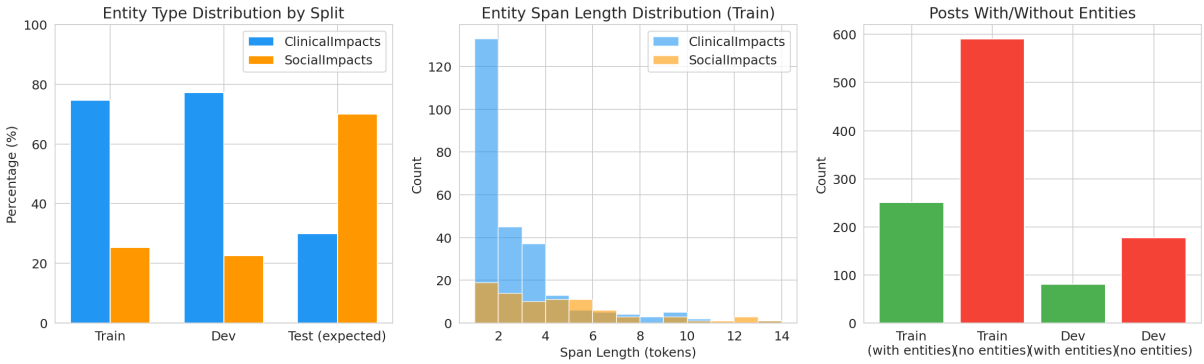

Figure 1: Reddit-Impacts 2.0 exploratory statistics. **Left:** Train/Dev are Clinical-heavy; Test flips to ≈70% Social. **Middle:** Social spans are longer and heavier-tailed than Clinical. **Right:** the majority of dev posts contain no entities at all, which makes a sentence-level presence head a useful auxiliary signal.

to Reddit-style noise and that aggressive cleaning removes genuine signal. Conversely, definition prompting and R-Drop only become their best selves in the preprocessed regime, because both rely on stable attention over a clean context. This asymmetry is exactly why treating preprocessing as an axis matters.

5 Methods

5.1 Model families

BiLSTM–CRF (sanity check). A frozen GloVe-300d embedding layer feeds a two-layer BiLSTM (2×256), a linear projection to five tags, and a CRF with Viterbi decoding at inference time.

GLiNER. We include two variants of GLiNER (Zaratiana et al., 2024): (a) fine-tuned on Reddit-Impacts with a 0.3 span-score threshold; and (b) a span-nested variant that enumerates nested candidate spans and applies overlap-aware decoding.

Domain-adapted encoders. SocBERT (Ji et al., 2022) is pre-trained on Reddit posts; Stress-RoBERTa (Amalq, 2022) is pre-trained on mental-health text. Both are token classifiers with a linear head.

DeBERTa-v3-large (main backbone). He et al. (2021)'s disentangled-attention encoder with a linear token classification head is our primary backbone. All advanced innovations build on top of it.

5.2 Contribution buckets

Our strategy is a roadmap of seven bucket-level contributions on top of DeBERTa. Each bucket is swept under $\mathrm{lr} \in \{2\mathrm{e}{-5}, 5\mathrm{e}{-5}\}$ in both regimes, and the best LR per family is reported.

5.2.1 Bucket 1: Loss, prompt, and multi-task ablations

Focal loss. We use the standard focal loss (Lin et al., 2017)

$$\mathcal{L}_{\text{focal}}(p_t) = -\,\alpha_t\,(1-p_t)^{\gamma}\,\log p_t,$$

with $\gamma = 2.0$ and a per-class weight $\alpha_{\text{Social}} = 1.5$, $\alpha_{\text{Clinical}} = \alpha_{\text{O}} = 1.0$ to push the model toward the under-represented Social class.

Definition prompting. We prepend a natural-language description of both entity types to every input, separated by a `[SEP]` token, and mask the prefix from the loss with -100 labels so that the definitions are available to attention but do not contribute gradient:

```
[CLS] Clinical Impacts are physical or
psychological consequences . . . Social
Impacts are social, occupational, legal
consequences . . . [SEP] <post> [SEP]
```

This is the single best-performing recipe in the project.

Multi-task auxiliary head. We share the DeBERTa encoder between the token NER head and a sentence-level binary head at `[CLS]` that predicts *entity presence* $y = \mathbf{1}[\text{sentence has any B/I tag}]$. The combined loss is $\mathcal{L} = \mathcal{L}_{\text{token}} + 0.3 \cdot \mathcal{L}_{\text{presence}}$.

5.2.2 Bucket 2: Advanced regularisation

Recall-Boost (O-class downweighting). Error analysis on the plain baseline showed recall trailing precision on the dev set, with 29 missed spans of which 13 were single-token implicit expressions. We respond by down-weighting the `O` class from 1.0 to 0.2 and up-weighting Social to 1.5, while keeping Clinical at 1.0. We combine this with the multi-task loss of the previous bucket.

R-Drop. We run each batch through the encoder twice with two independent dropout masks and add a symmetric KL penalty between the two output distributions:

$$\mathcal{L} = \tfrac{1}{2}(\mathrm{CE}(p_1) + \mathrm{CE}(p_2)) + \alpha \cdot \mathrm{KL}_{\mathrm{sym}}(p_1 \,\|\, p_2),$$

with $\alpha = 1.0$ applied only on non-masked tokens. R-Drop introduces no new parameters.

FGM + SWA. Per training step we generate an adversarial perturbation $w \leftarrow w + \varepsilon \cdot g/\|g\|$ on the `word_embeddings` layer with $\varepsilon = 0.5$ (Miyato et al., 2017), take a second forward pass, sum the gradients from the clean and adversarial passes, and step the optimiser. After epoch 10 we compute a stochastic weight average (Izmailov et al., 2018) over the remaining epoch checkpoints and use the averaged weights as the final model.

5.2.3 Bucket 3: Preprocessing axis

See §4.

5.2.4 Bucket 4: Structured prediction pipelines

Beyond token-level classification, we implement four structured variants:

Hierarchical DeBERTa. A `[CLS]` sentence classifier gates a multi-task NER: only sentences whose sentence head predicts "has-entity" are routed to the token head. During training we keep every positive sentence and only 10% of the no-impact sentences, so the token head is trained on a more balanced distribution.

Two-Step Impact Pipeline. Stage 1 is a binary span extractor ("is this a substance-use impact?"); Stage 2 classifies the extracted span as Clinical or Social using a separate head. The two stages are trained independently.

Sentence + Token Hierarchy. A single model with two heads trained jointly: one for sentence presence, one for token BIO.

Span-nested GLiNER. A span scorer with explicit nested-span enumeration and overlap-aware decoding at inference.

5.2.5 Bucket 5: Domain-adapted backbones

SocBERT (Reddit-domain) and StressRoBERTa (mental-health) are swapped in under the transformer token-classifier template while keeping the same optimiser, LR sweep, and training schedule.

5.2.6 Bucket 6: Weight-space model soup

For a given family we fine-tune at every $(\mathrm{lr}, \mathrm{seed})$ configuration and average the weights of the top-K dev-F1 checkpoints: $\bar{\theta} = \frac{1}{K}\sum_k \theta_k$. This is only defined for same-architecture runs, so GLiNER, BiLSTM–CRF, and every pipeline family above are *not* soupable.

5.2.7 Bucket 7: Prediction-space ensemble search

We run two complementary exhaustive searches, both over size-2–5 subsets:

- **Majority vote (cross-regime).** For each regime we take the top-7 relaxed-F1 runs and the top-7 strict-F1 runs, yielding up to 28 unique candidates. We then exhaustively evaluate all 122,409 size-2–5 subsets, picking the BIO tag for each token by hard majority. Because the voting operates on final BIO sequences, structured pipelines and GLiNER variants can participate.
- **Probability average (raw only).** Seven candidates, 112 size-2–5 subsets. Averaging is performed on per-token class distributions after re-aligning to the original token grid, so GLiNER span models and every structured pipeline—which do not expose comparable per-token tensors—are excluded from the candidate pool.

6 Experiments

Setup. All experiments use AdamW with a linear warmup of 10% of total steps and linear decay. DeBERTa family trains with batch size 16 for 12 epochs, dropout 0.1, max sequence length 512, and gradient clipping at 1.0. BiLSTM–CRF trains for 15 epochs with SGD 0.01. GLiNER fine-tunes with the authors' default schedule.

Protocol. Innovations are compared *within* a regime first; the only stage that mixes regimes is the cross-regime ensemble search (§5.2.7). All tables below report dev-set F1; no test labels are used for selection.

6.1 Baseline model comparison

Table 1 compares the five model families under the raw-vs-preprocessed axis. BiLSTM–CRF and GLiNER gain from preprocessing; plain DeBERTa is stronger on raw text. Both domain-adapted encoders sit well below DeBERTa even in

Model	Raw	Prep
BiLSTM–CRF	0.251	0.296
GLiNER (fine-tuned)	0.463	0.524
SocBERT (Reddit)	–	0.404
StressRoBERTa (mental-health)	–	0.482
DeBERTa-v3-large baseline	**0.595**	0.545

Table 1: Model family comparison, relaxed F1 on dev. GLiNER zero-shot scores 0.320 and is omitted for brevity.

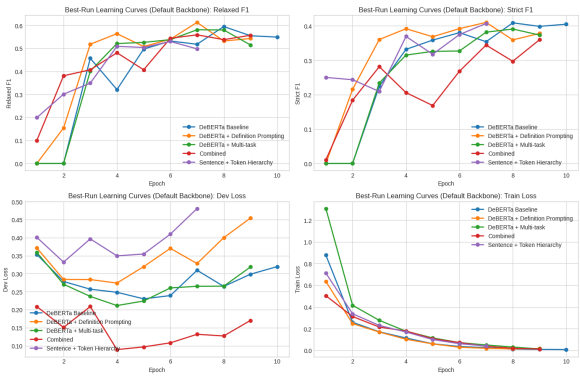

Figure 2: Best-run learning curves. **Top row:** Definition prompting (orange) tops both relaxed and strict F1 from epoch 4 onward, the clearest single-recipe win. **Bottom row:** dev loss separates recipes even when F1 curves look close; the Sentence + Token Hierarchy drifts up after epoch 6, signalling mild overfit on this small training set.

their own preferred regime: a Reddit- or mental-health-adapted pre-training corpus does not translate into a stronger BIO tagger for impact extraction.

6.2 Per-bucket ablations

Table 2 reports the per-bucket effects on top of the DeBERTa baseline, with relaxed/strict F1 shown in both regimes. The pattern is striking: *only* definition prompting improves on the baseline in either regime. Focal loss, multi-task, recall-boost, R-Drop, and FGM + SWA all sit below their own-regime baseline when measured in isolation.

The learning-curve panel in Figure 2 makes the effect visible: definition prompting and the plain baseline plateau at ≈ 0.60 relaxed F1, while other recipes oscillate around or below it. Dev loss is a noisier signal than F1 and sometimes separates recipes that F1 treats as similar (the Sentence + Token Hierarchy variant drifts up after epoch 6, which the raw F1 number does not show).

6.3 Structured prediction pipelines

Table 3 compares the four structured variants. Hierarchical DeBERTa is the clear winner and becomes the anchor of our final ensemble. Strict F1 on the hierarchical pipeline actually goes up under preprocessing even though relaxed F1 goes down: the sentence-gating head tightens predicted boundaries when the token head is working with cleaner input. The Two-Step Impact Pipeline trades recall for precision and reaches the highest strict F1 of any single-model recipe (0.404).

6.4 Model soup

The best soup averages the top-3 checkpoints of the Recall-Boost seed-42 LR sweep and reaches relaxed F1 $= 0.595$, i.e. exactly the plain DeBERTa baseline, with strict F1 0.331 (the strict number is worse). Soup recovers some of the damage from the Recall-Boost experiment but does not exceed the un-souped baseline. This is consistent with the findings of Wortsman et al. (2022): soup helps when the underlying recipe is already competitive, and does not rescue weaker recipes.

6.5 Ensemble search

Table 4 shows the top ensemble sizes from the cross-regime majority-vote search, plus the best probability-average result. The four-model majority ensemble reaches relaxed F1 $=$ **0.635**, a $+0.022$ jump over the best single model and $+0.025$ over the published baseline on the same dev split. The strict-F1 optimum is a different five-model subset at 0.518. Hard voting consistently beats soft averaging on this benchmark (0.635 vs 0.614), which is a reversal of the usual deep-learning preference for logit averaging.

The winning four-model subset pairs the Hierarchical DeBERTa pipeline as an anchor with three complementary DeBERTa recipes drawn from both preprocessing regimes: a definition-prompting run from the preprocessed regime and two raw-regime runs with different regularisation objectives. The ensemble benefits from both *preprocessing diversity* (members trained on different input distributions) and *objective diversity* (definition prompting, multi-task, sentence gating), which is where the $+0.022$ headroom over the best single model comes from.

6.6 Full ablation

Table 5 presents the full LR-selected ablation in one view. BiLSTM–CRF sits at

Recipe	Raw (R / S) Relaxed	Raw (R / S) Strict	Preprocessed (R / S) Relaxed	Preprocessed (R / S) Strict
DeBERTa baseline	0.595	0.409	0.545	0.421
+ Focal loss	0.541	0.341	0.538	0.358
+ Multi-task	0.581	0.382	0.550	0.394
+ Definition prompting	0.575	0.350	**0.613**	**0.410**
+ Recall-Boost (seed 42)	0.582	0.312	0.560	0.330
+ R-Drop (seed 123)	0.561	0.360	0.577	0.380
+ FGM + SWA (seed 42)	0.573	0.356	0.547	0.256

Table 2: Per-bucket effect on top of the DeBERTa baseline. All numbers are dev F1; best LR per family is selected from $\{2e{-}5, 5e{-}5\}$ within each regime. Only definition prompting improves on the baseline, and only under preprocessing.

Pipeline	Raw (R / S)	Prep (R / S)
Hierarchical DeBERTa	**0.604** / 0.372	0.559 / 0.383
Two-Step Impact	–	0.534 / 0.404
Sentence + Token	–	0.530 / 0.375
Span-nested GLiNER	–	0.443 / 0.491

Table 3: Structured prediction pipelines (dev F1). Only Hierarchical DeBERTa was run under both regimes; the others are preprocessed only.

Ensemble	Relaxed	Strict	Prec.	Rec.
2 models (majority)	0.623	0.465	0.586	0.664
3 models (majority)	0.621	0.507	0.638	0.605
4 models (majority)	**0.635**	0.509	0.648	0.623
5 models (majority)	0.631	**0.518**	0.670	0.597
Prob. avg. (raw, 7 cand.)	0.614	0.431	0.653	0.580

Table 4: Ensemble search. Cross-regime majority vote across 122,409 size-2–5 subsets; probability average across 112 size-2–5 subsets in the raw regime only (span/pipeline families are excluded because they do not emit per-token distributions).

Model	Relaxed	Strict	LR
BiLSTM-CRF	0.296	0.256	5e-4
GLiNER (fine-tuned)	0.524	–	2e-5
Span-nested GLiNER	0.443	–	1e-5
SocBERT	0.404	0.277	5e-5
StressRoBERTa	0.482	0.270	2e-5
DeBERTa baseline	0.595	0.409	2e-5
+ Focal loss	0.541	0.341	2e-5
+ Multi-task	0.581	0.382	5e-5
+ Definition prompting	**0.613**	0.410	5e-5
Hierarchical DeBERTa	0.604	0.372	2e-5
Two-Step Impact	0.534	0.404	2e-5
Sentence + Token	0.530	0.374	5e-5
Recall-Boost (s42)	0.582	0.312	2e-5
R-Drop (s123)	0.577	0.380	2e-5
FGM + SWA (s42)	0.573	0.356	2e-5
Model soup (recall-boost s42)	0.595	0.331	2e-5
4-model majority ensemble	**0.635**	0.509	–
GPT-4o 3-shot (reported)	*0.440*	–	–
DeBERTa published SOTA	*0.610*	–	–

Table 5: Full LR-selected ablation (dev F1). Strict F1 for GLiNER variants is not exposed by the toolkit.

0.296; domain-adapted encoders underperform the general-purpose DeBERTa; the only recipe that improves on the baseline alone is definition prompting; the four-model ensemble sits clearly above the published SOTA on the same split.

7 Error Analysis

We ran a targeted error analysis on the best relaxed single model (definition prompting, preprocessed regime, $\mathrm{lr} = 5e{-}5$, relaxed F1 $= 0.613$, strict F1 $= 0.410$). Four categories dominate:

Boundary errors. The ≈ 0.20 gap between relaxed and strict F1 is almost entirely attributable to partial span matches on multi-word Social entities. Phrases such as "charged with disorderly conduct" are typed correctly but the predicted span misses the "charged with" prefix.

Implicit false negatives. Single-token implicit Clinical expressions ("hell", "broken", "numb") and long-tail colloquial phrases are the dominant category of missed spans. These spans do not contain any clinical vocabulary and the model falls back to the "no-entity" prior.

Negation handling. Negated phrases are mostly rejected correctly; type confusion between Clinical and Social is rare.

Third-person references. The model over-rejects spans that describe someone else's impacts rather than the poster's, despite no explicit training signal for this distinction.

These categories suggest two concrete future-work directions: boundary-aware decoding for the first category, and targeted synthetic augmentation for the second.

8 Discussion and Key Insights

Our headline finding is that *training-objective diversity* is a much stronger lever on this benchmark than individual-model polish. A four-model majority-vote ensemble of relatively ordinary recipes beats any single one of them by $+0.022$ relaxed F1. Three secondary observations are worth calling out:

LR sweeps are non-negotiable. The best LR is family-specific ($2\mathrm{e}{-5}$ for the plain baseline, $5\mathrm{e}{-5}$ for definition prompting and multi-task). Running a single global LR would have reversed the ordering between at least three recipes and hidden definition prompting's improvement.

Domain adaptation is not a free lunch. SocBERT and StressRoBERTa, both plausible a priori, underperform general-purpose DeBERTa by ≈ 0.09–0.19 relaxed F1. For a task whose labels are semantic categories defined at annotation time, generic language understanding appears to matter more than register-specific pre-training.

Majority vote beats probability average. On our dev set hard voting is $+0.021$ better than soft averaging (0.635 vs 0.614). We hypothesise that the most confident recipes are also the most overconfident: their logit mass dominates the average and propagates their particular mistakes, whereas a hard vote caps each member's influence at one. On a small, noisy benchmark with decorrelated errors, this cap is worth more than the information lost by discretising the output.

9 Limitations

Dev-only evaluation. We report dev-set performance throughout. Test labels were not available during the study, and the dev split itself is only 258 documents, so a confidence interval on relaxed F1 is wide. Given the $75\% \rightarrow 70\%$ label prior flip, test numbers may well shift from what we report; the system rankings we report are meaningful *within* the dev distribution but do not guarantee test-set generalisation.

No cross-post context. Our models see one post at a time. User history within a subreddit is a potentially strong signal, particularly for disambiguating third-person references, and we do not exploit it.

Ensemble cost. The best system runs four models at inference time and pays a $\approx 4\times$ latency penalty relative to a single-model deployment. In practice the hierarchical pipeline at 0.604 captures ≈ 0.03 below the ensemble at a single-model cost.

Coverage of domain encoders. Budget constraints restricted SocBERT and StressRoBERTa to the preprocessed regime only. It is possible—but in our view unlikely given the raw/prep ordering of the other transformers—that a raw run of either encoder would close the gap to DeBERTa.

10 Conclusion

We presented a systematic study of the Reddit-Impacts impact-extraction task, organised as seven contribution buckets layered on top of a DeBERTa backbone under a consistent LR sweep and a paired raw-vs-preprocessed protocol. Our best single model, definition prompting, reaches relaxed F1 $= 0.613$. Our best system, a four-model majority-vote ensemble, reaches 0.635 and exceeds the published baseline by $+0.025$ on the same dev split. The pattern across buckets is consistent: no single regularisation trick beats a well-tuned DeBERTa, and the headroom comes from combining decorrelated recipes via hard voting. We see the ≈ 0.18 gap to the human ceiling as coming mainly from boundary decoding and implicit-expression recall, and view these as the most promising future-work directions.

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A Hyperparameters and Training Setup

All DeBERTa-family runs use AdamW with $\beta = (0.9, 0.999)$, weight decay 0.01, batch size 16, max sequence length 512, dropout 0.1, linear warmup over 10% of total steps, and linear decay. LR is swept from $\{2\mathrm{e}{-5}, 5\mathrm{e}{-5}\}$. Training runs for 12 epochs unless otherwise noted; SWA averaging starts after epoch 10. BiLSTM–CRF trains for 15 epochs with SGD 0.01. GLiNER fine-tuning follows the default schedule of Zaratiana et al. (2024) with a 0.3 span-score threshold.

B Reproducibility

All experiments use PyTorch with mixed-precision training on a single NVIDIA A100 GPU. The cross-regime ensemble search runs as a pure CPU post-processing step on cached BIO sequences from each candidate and takes roughly 45 minutes for all $122{,}409$ size-2–5 subsets. The probability-average search takes ≈ 5 minutes on the 112 candidate subsets.